

Daniel W. Cranston

dcransto@gmail.com

<http://www.graphcoloringmethods.com/writing.html>

Research Interests

Graph theory and algorithm design; in particular, structural and extremal graph theory and graph coloring.

Academic Appointments

- 2020–2025 **Virginia Commonwealth University**; Associate Professor of Computer Science (with tenure).
2015–2020 **Virginia Commonwealth University**; Associate Professor of Mathematics (with tenure).
2016–2017 **Johns Hopkins University**; Visiting Associate Professor of Applied Math and Statistics.
2009–2015 **Virginia Commonwealth University**; Assistant Professor of Mathematics.
2007–2009 **Center for Discrete Math and Theoretical Computer Science (DIMACS) and Bell Labs**; Rutgers University and Murray Hill, N.J. Postdoctoral Fellow.

Education

- 1999–2007 **University of Illinois at Urbana-Champaign**:
Ph.D. in Computer Science, May 2007. Advisor: Douglas West.
Thesis: Coloring and Labeling Problems on Graphs.
M.S. in Computer Science, January 2004. Advisor: Jeff Erickson.
M.S. in Applied Mathematics, Optimization and Algorithms, January 2001.
1996–1999 **Greenville College**:
B.A. Major in Mathematics, magna cum laude, May 1999. Honors program.

Editorial Work

- 2025–present Editor, Electronic Journal of Combinatorics
2018–present Editorial Board, Journal of Graph Theory
2019–present Editorial Board, Discussiones Mathematicae Graph Theory
2018–2025 Managing Editor, Australasian Journal of Combinatorics
2023 Guest Editor (with H.A. Kierstead and B. Rabern)
Special Issue of Discrete Mathematics in Honour of Landon Rabern
Volume 346, Issue 11, November 2023

Awards and Honors

- 2024–2025 Engineering Expo 3rd Place (1 of 2) in Computer Science, out of 49
Community Fridges Usage Data Acquisition: Team Adviser
2023–2024 Engineering Expo 3rd Place in Multidisciplinary, out of 20; Sternheimer Winner (1 of 4)
Little Ram Pantries Usage and Inventory Acquisition and Visualization: Team Adviser
2022–2023 Engineering Expo Overall 1st Place (1 of 2), out of 92; Sternheimer Winner (1 of 4)
Web App for Assessing Prewriting Skill in Children: Team Adviser
2014, 2017, Nominated by Math Department for VCU Humanities and Sciences
2018, 2019 Excellence in Scholarship Award (one nomination per department)
2009–2010 Fellow of AMS Project NExT (Professional Development Program)

Publications

H-index 22, i10-index 45, and 1756 citations in Google Scholar as of 19 August 2026. Copies of these papers can be downloaded by clicking on the titles or from https://arxiv.org/a/cranston_d_1.html.

Each author below marked with * was a student when the research was performed.

Books

- [B1] Daniel W. Cranston, *Graph Coloring Methods*. viii+453 pages, July 2024. (A graduate textbook.) Available for free download at graphcoloringmethods.com

Papers Submitted for Publication

- [88] Daniel W. Cranston, Jiaao Li, Bo Su*, Zhouningxin Wang, and Ningyan Xu*. *Reconfiguration of Nowhere-zero Flows*.
- [87] Wouter Cames van Batenburg, Daniel W. Cranston, and František Kardoš. *Disjoint Correspondence Colorings for K_5 -Minor-free Graphs*.
- [86] Daniel W. Cranston. *Progress on Albertson's Conjecture*.
- [85] Daniel W. Cranston and Reem Mahmoud. *Equitably Coloring Planar and Outerplanar Graphs*.
- [84] Stijn Cambie, Wouter Cames van Batenburg, Daniel W. Cranston, Jan van den Jeuel, and Ross J. Kang. *Reconfiguration of List Colourings*.
- [83] Daniel W. Cranston, Jiaao Li, Zhouningxin Wang, and Chunyan Wei*. *Planar Graphs with Homomorphisms to the 9-cycle*.

Refereed Journal Publications

- [82] Nicolas Bousquet and Daniel W. Cranston. *A Linear Kernel for Independent Set Reconfiguration in Planar Graphs*. SIAM J. Discrete Math. To appear.
- [81] Daniel W. Cranston, Jiaao Li, Bo Su*, Zhouningxin Wang, and Chunyan Wei*. *Orientations of 10-Edge-Connected Planar Multigraphs and Applications*. Journal of Graph Theory. 19 Aug. 2026. 25pp.
- [80] Stijn Cambie, Wouter Cames van Batenburg, and Daniel W. Cranston. *Sharp Bounds on Lengths of Linear Recolouring Sequences*. Electronic Journal of Combinatorics. Vol. 33(1), 2026, #P1.18. 20 pp.
- [79] Daniel W. Cranston. *10-list Recoloring of Planar Graphs*. European Journal of Combinatorics. Vol. 130, December 2025, Article 104190, 20 pp.
- [78] Daniel W. Cranston and Evelyne Smith-Roberge. *List Packing and Correspondence Packing of Planar Graphs*. Journal of Graph Theory. Vol. 109(3), July 2025, pp. 339–352.
- [77] Daniel W. Cranston and Reem Mahmoud*. *Equitable Coloring in 1-Planar Graphs*. Discrete Math. Vol. 348(2), Feb. 2025, 114286.
- [76] Daniel W. Cranston and Chun-Hung Liu. *Proper Conflict-free Coloring of Graphs with Large Maximum Degree*. SIAM Journal on Discrete Math. Vol. 38(4), 2024, pp. 3004–3027.
- [75] Daniel W. Cranston and Carl Feghali. *Kempe Classes and Almost Bipartite Graphs*. Discrete Applied Math. Vol. 357, 15 November 2024, pp. 94–98.
- [74] Daniel W. Cranston and Gexin Yu. *Cliques in Squares of Graphs with Maximum Average Degree less than 4*. Journal of Graph Theory. Vol. 107(3). July 2024. pp. 559–577.
- [73] Daniel W. Cranston. *Bounding Clique Size in Squares of Planar Graphs*. European Journal of Combinatorics. Vol. 120, August 2024, Article 103960, 9 pp.
- [72] Daniel W. Cranston and Reem Mahmoud*. *Kempe Equivalent List Colorings*. Combinatorica. Vol. 44, 2024, pp. 125–153.
- [71] Daniel W. Cranston and Reem Mahmoud*. *5-Coloring Reconfiguration of Planar Graphs with No Short Odd Cycles*. Journal of Graph Theory. Vol. 105(4). April 2024. pp. 670–679.
- [70] Daniel W. Cranston. *Odd-Sum Colorings of Planar Graphs*. Discrete Applied Math. Vol. 342, 15 January 2024, pp. 82–88.
- [69] Stijn Cambie, Wouter Cames van Batenburg, and Daniel W. Cranston. *Optimally Reconfiguring List and Correspondence Colourings*. European J. Comb. Vol. 115, Jan. 2024, Article 103798, 20 pp.

- [68] Daniel W. Cranston. *Odd Colorings of Sparse Graphs*. Journal of Combinatorics. Vol. 15(4), 2024, pp. 439–450.
- [67] Daniel W. Cranston and Hudson LaFayette*. *The t -Tone Chromatic Number of Classes of Sparse Graphs*. Australasian J. Combinatorics. Vol. 86(3), 2023, pp. 458–476.
- [66] Daniel W. Cranston. *Coloring, List Coloring, and Painting Squares of Graphs (and other related problems)*. Electronic Journal of Combinatorics. Vol. 30(2), 21 April 2023, Dynamic Survey #25.
- [65] Daniel W. Cranston, Michael Lafferty*, and Zi-Xia Song. *A Note on Odd Colorings of 1-Planar Graphs*. Discrete Applied Math. Vol. 330, 15 May 2023, pp. 112–117.
- [64] Daniel W. Cranston, Jiaxi Nie*, Jacques Verstraëte, and Alexandra Wesolek*. *On Asymptotic Packing of Geometric Graphs*. Discrete Applied Math. Vol. 322, 15 December 2022, pp. 142–152.
- [63] Daniel W. Cranston. *Kempe Equivalent List Edge-Colorings of Planar Graphs*. Discrete Math (Special Issue in Honour of Landon Rabern). Vol. 346(11), Nov. 2023, 113180.
- [62] Daniel W. Cranston, Bernard Lidický, Xiaonan Liu*, and Abhinav Shantanam*. *Planar Turán Numbers of Cycles: A Counterexample*. Electronic Journal of Combinatorics. Vol. 29(3), 2022, #P3.31.
- [61] Daniel W. Cranston. *Strong Edge-Coloring of Cubic Bipartite Graphs: A Counterexample*. Discrete Applied Math. Vol. 321, 15 November 2022, pp. 258–260.
- [60] Daniel W. Cranston. *List-Recoloring of Sparse Graphs*. European Journal of Combinatorics. Vol. 105, October 2022, 103562.
- [59] Daniel W. Cranston and Reem Mahmoud*. *In Most 6-regular Toroidal Graphs All 5-colorings are Kempe Equivalent*. European Journal of Combinatorics. Vol. 104, August 2022, 103532.
- [58] Daniel W. Cranston and Matthew P. Yancey. *Vertex Partitions into an Independent Set and a Forest with Each Component Small*. SIAM Journal on Discrete Math. Vol. 35(3), 2021, pp. 1769–1791.
- [57] Daniel W. Cranston, Hudson Lafayette*, and Landon Rabern. *Coloring (P_5, gem) -free graphs with $\Delta - 1$ colors*. Journal of Graph Theory. Vol. 101(4), December 2022, pp. 633–642.
- [56] Daniel W. Cranston and Matthew P. Yancey. *Sparse Graphs are Near-bipartite*. SIAM Journal on Discrete Math. Vol. 34(3), 2020, pp. 1725–1768.
- [55] Daniel W. Cranston and Jiaao Li. *Circular Flows in Planar Graphs*. SIAM Journal on Discrete Math. Vol. 34(1), 2020, pp. 497–519.
- [54] Ilkyoo Choi, Daniel W. Cranston, and Théo Pierron*. *Degeneracy and Colorings of Squares of Planar Graphs without 4-Cycles*. Combinatorica. Vol. 40(5), 2020, pp. 625–653.
- [53] Neal Bushaw and Daniel W. Cranston. *A Note on Bootstrap Percolation Thresholds in Plane Tilings using Regular Polygons*. Australasian J. Combinatorics. Vol. 74(3), 2019, pp. 486–497.
- [52] Daniel W. Cranston and Landon Rabern. *The Hilton–Zhao Conjecture is True for Graphs with Maximum Degree 4*. SIAM Journal on Discrete Math. 33(3), 2019, pp. 1228–1241.
- [51] Daniel W. Cranston. *A Characterization of $(4,2)$ -Choosable Graphs*. Journal of Graph Theory. Vol. 92(4), December 2019, pp. 460–487.
- [50] Daniel W. Cranston. *Acyclic Edge-coloring of Planar Graphs: Δ Colors Suffice when Δ is Large*. SIAM Journal on Discrete Math. Vol. 33(2), 2019, pp. 614–628.
- [49] Daniel W. Cranston. *Proper Distinguishing Colorings with Few Colors for Graphs with Girth at Least 5*. Electronic Journal of Combinatorics. Vol. 25(3), 2018, #P3.5.
- [48] Marthe Bonamy, Daniel W. Cranston, and Luke Postle. *Planar Graphs of Girth at least Five are Square $(\Delta + 2)$ -Choosable*. Journal of Combinatorial Theory, Series B. Vol. 134, 2019, pp. 218–238. **Among 25 “most downloaded” JCTB papers (March 2019).**
- [47] Daniel W. Cranston and Landon Rabern. *Planar Graphs are $9/2$ -colorable*. Journal of Combinatorial Theory, Series B. Vol. 133, 2018, pp. 32–45. **Among 24 “most downloaded” JCTB papers (from December 2018 through February 2019).**
- [46] Daniel W. Cranston and Landon Rabern. *Beyond Degree Choosability*. Electronic Journal of Combinatorics. Vol. 24(3), 2017, #P3.29.
- [45] Daniel W. Cranston and Landon Rabern. *Short Fans and the $5/6$ Bound for Line Graphs*. SIAM Journal on Discrete Math. Vol. 31(3), 2017, pp. 2039–2063.

- [44] Daniel W. Cranston and Landon Rabern. *List-coloring Claw-free Graphs with $\Delta - 1$ colors*. SIAM Journal on Discrete Math. Vol. 31(2), 2017, pp. 726–748.
- [43] Daniel W. Cranston and Landon Rabern. *Edge Lower Bounds for List Critical Graphs, via Discharging*. Combinatorica. Vol. 38(5), October 2018, pp. 1045–1065.
- [42] Daniel W. Cranston and Landon Rabern. *Subcubic Edge Chromatic Critical Graphs have Many Edges*. Journal of Graph Theory. Vol. 86(1), September 2017, pp. 122–136.
- [41] Daniel W. Cranston and Douglas B. West. *An Introduction to the Discharging Method via Graph Coloring*. Discrete Math. Vol. 340, no. 4, April 2017, pp. 766–793.
Extended version: *A Guide to the Discharging Method*. At: <https://arxiv.org/abs/1306.4434v1>
- [40] Daniel W. Cranston and Robert Jaeger*. *Coloring Squares of Planar Graphs with no 4-cycles and no 5-cycles*. Journal of Graph Theory. Vol. 85(4), August 2017, pp. 721–737.
- [39] Daniel W. Cranston, Luke Postle, Chenxiao Xue*, and Carl Yerger. *Modified Linear Programming and Class 0 Bounds for Graph Pebbling*. J. Combinatorial Optimization. 34(1), July 2017, pp. 114–132.
- [38] Daniel W. Cranston and Landon Rabern. *Planar Graphs have Independence Ratio at least $3/13$* . Electronic Journal of Combinatorics. Vol. 23(3), 2016, #P3.45.
- [37] Daniel W. Cranston and Landon Rabern. *Painting Squares in $\Delta^2 - 1$ Shades*. Electronic Journal of Combinatorics. Vol. 23(2), 2016, #P2.50.
- [36] Daniel W. Cranston and Landon Rabern. *Graphs with $\chi = \Delta$ have Big Cliques*. SIAM Journal on Discrete Math. Vol. 29(4), 2015, pp. 1792–1814.
- [35] Daniel W. Cranston and Landon Rabern. *The Fractional Chromatic Number of the Plane*. Combinatorica. Vol. 37(5), October 2017, pp. 837–861.
- [34] Daniel W. Cranston and Landon Rabern. *A Note on Coloring Vertex-transitive Graphs*. Electronic Journal of Combinatorics. Vol. 22(2), 2015, #P2.1.
- [33] Daniel W. Cranston and Landon Rabern. *Brooks’ Theorem and Beyond*. J. Graph Theory. Vol. 80(3), November 2015, pp. 199–225.
- [32] Daniel W. Cranston and Landon Rabern. *Coloring a graph with $\Delta - 1$ colors: Conjectures Equivalent to the Borodin-Kostochka Conjecture that Appear Weaker*. European J. Comb. Vol. 44, Part A, February 2015, pp. 23–42.
- [31] Daniel W. Cranston, Yuchang Liang*, and Xuding Zhu. *Regular Graphs of Odd Degree are Antimagic*. Journal of Graph Theory. Vol. 80(1), September 2015, pp. 28–33.
- [30] Daniel W. Cranston, Rok Erman, and Riste Škrekovski. *Choosability of the Square of a Planar Graph with Maximum Degree Four*. Australasian Journal of Combinatorics. Vol. 59(1), June 2014, pp. 86–97.
- [29] Daniel W. Cranston and Riste Škrekovski. *Sufficient Sparseness Conditions for G^2 to be $(\Delta + 1)$ -choosable when $\Delta \geq 5$* . Discrete Applied Math. Vol. 162(10), January 2014, pp. 167–176.
- [28] Daniel W. Cranston and Candace M. Kent. *On the Boundedness of Positive Solutions of the Difference Equation $x_n = \max \left\{ \frac{A_{n-1}^1}{x_{n-1}}, \frac{A_{n-1}^2}{x_{n-2}}, \dots, \frac{A_{n-1}^t}{x_{n-t}} \right\}$ with Periodic Parameters*. Applied Mathematics and Computation. Vol. 221, 2013, pp. 144–151.
- [27] Daniel W. Cranston and Suil O. *Hamiltonicity in Connected Regular Graphs*. Information Processing Letters Vol. 113, 2013, pp. 858–860.
- [26] Daniel W. Cranston, Sogol Jahanbekam*, and Douglas B. West. *1,2,3-Conjecture and 1,2-Conjecture for Sparse Graphs*. Discussiones Mathematicae Graph Theory. Vol. 34(4), 2014, pp. 769–799.
- [25] Daniel W. Cranston, Jaehoon Kim, and William B. Kinnersley. *New Results in t -tone Coloring*. Electronic Journal of Combinatorics. Vol. 20(2), 2013, #P17.
- [24] Daniel W. Cranston and Landon Rabern. *Coloring Claw-free Graphs with $\Delta - 1$ Colors*. SIAM Journal on Discrete Math. Vol. 27(1), 2013, pp. 534–549.
- [23] Daniel W. Cranston, William B. Kinnersley, Suil O, and Douglas B. West. *Game Matching Number of Graphs*. Discrete Applied Math. Vol. 161(13–14), 2013, pp. 1828–1836.
- [22] Daniel W. Cranston, William B. Kinnersley, Kevin G. Milans, Gregory J. Puleo, and Douglas B. West. *Chain-making Games in Grid-like Posets*. Journal of Combinatorics. Vol. 3(4), 2012, pp. 633–650.

- [21] Jane Butterfield, Daniel W. Cranston, Gregory J. Puleo, Douglas B. West, and Reza Zamani. *Revolutionaries and spies: Spy-good and Spy-bad Graphs*. Theoret. Comput. Sci. Vol. 463, 2012, pp. 35–53.
- [20] Daniel W. Cranston, Anja Pruchnewski, Zsolt Tuza, and Margit Voigt. *List-colorings of K_5 -minor-free Graphs with Special List Assignments*. J. of Graph Theory. Vol. 71(1), September 2012, pp. 18–30.
- [19] Daniel W. Cranston, Clifford Smyth, and Douglas B. West. *Revolutionaries and Spies on Trees and Unicyclic Graphs*. Journal of Combinatorics. Vol. 3(2), 2012, pp. 195–206.
- [18] Daniel W. Cranston, Nitish Korula, Tim LeSaulnier, Kevin Milans, Chris Stocker, Jennifer Vandembussche, and Douglas B. West. *Overlap Number of Graphs*. Journal of Graph Theory. Vol. 70(1), May 2012, pp. 10–28.
- [17] Daniel W. Cranston and Gexin Yu. *Linear Choosability of Sparse Graphs*. Discrete Math. Vol. 311, no. 17, 6 September 2011, pp. 1910–1917.
- [16] Daniel W. Cranston, Seog-Jin Kim, and Gexin Yu. *Injective Colorings of Graphs with Low Average Degree*. Algorithmica. Vol. 60(3), July 2011, pp. 553–568.
- [15] Daniel W. Cranston, Seog-Jin Kim, and Gexin Yu. *Injective Colorings of Sparse Graphs*. Discrete Math. Vol. 310, no. 21, 6 November 2010, pp. 2965–2973.
- [14] Yuehua Bu, Daniel W. Cranston, Mickaël Montassier, André Raspaud, and Weifan Wang. *Star Coloring of Sparse Graphs*. Journal of Graph Theory. Vol. 62(3), November 2009, pp. 201–219.
- [13] Daniel W. Cranston and Gexin Yu. *A New Lower Bound on the Density of Vertex Identifying Codes for the Infinite Hexagonal Grid*. Electronic Journal of Combinatorics. Vol. 16(1), 2009, #R113.
- [12] Daniel W. Cranston. *Multigraphs with $\Delta \geq 3$ are Totally- $(2\Delta - 1)$ -Choosable*. Graphs and Combinatorics. Vol. 25(1), May 2009, pp. 35–40.
- [11] Daniel W. Cranston. *Edge-choosability and Total-choosability of Planar Graphs with no Adjacent 3-cycles*. Discussiones Mathematicae Graph Theory. Vol. 29(1), 2009, pp. 163–178.
- [10] Daniel W. Cranston and Douglas B. West. *Classes of 3-regular Graphs that are $(7,2)$ -edge-choosable*. SIAM Journal on Discrete Math. Vol. 23(2), April 2009, pp. 872–881.
- [9] Charles Mullins and Daniel W. Cranston. *Research at ASMSA Based on the DIMACS Biomath Program*. DIMACS Series in Discrete Math and Theoretical Computer Science. Vol. 76, pp. 221–226.
- [8] Michael O. Albertson, Daniel W. Cranston, and Jacob Fox. *Crossings, Colorings, and Cliques*. Electronic Journal of Combinatorics. Vol. 16(1), 2009, #R45.
- [7] Daniel W. Cranston. *Regular Bipartite Graphs are Antimagic*. J. of Graph Theory. Vol. 60, March 2009, pp. 173–182. **Among 10 “most cited” recent JGT publications (as of August 2011).**
- [6] Wenjie He, Lingmin Zhang, Daniel Cranston, Yufa Shen, Guoping Zheng. *Choice Number of Complete Multipartite Graphs $K_{4,3*2,2*(k-6),1*3}$ and $K_{3*3,2*(k-5),1*2}$* . Discrete Math. Vol. 308, no. 23, 6 December 2008, pp. 5871–5877.
- [5] Daniel W. Cranston. *Nomadic Decompositions of Complete Bidirected Graphs*. Discrete Math. Vol. 308, no. 17, 6 September 2008, pp. 3982–3985.
- [4] David P. Bunde, Erin W. Chambers, Daniel W. Cranston, Kevin Milans, and Douglas B. West. *Pebbling and Optimal Pebbling in Graphs*. Journal of Graph Theory. Vol. 57, March 2008, pp. 215–238.
- [3] Daniel W. Cranston and Seog-Jin Kim. *List-coloring the Square of a Subcubic Graph*. Journal of Graph Theory. Vol. 57, January 2008, pp. 65–87.
- [2] Daniel W. Cranston, I. Hal Sudborough, and Douglas B. West. *Short Proofs for Cut-and-Paste Sorting of Permutations*. Discrete Math. Vol. 307, no. 22, 28 October 2007, pp. 2866–2870.
- [1] Daniel W. Cranston. *Strong Edge-coloring of Graphs with Maximum Degree 4 using 22 Colors*. Discrete Math. Vol. 306, no. 21, 6 November 2006, pp. 2772–2778.

Conference Papers

- [C3] Nicolas Bousquet and Daniel W. Cranston. *A Linear Kernel for Independent Set Reconfiguration in Planar Graphs*. Symposium on Theoretical Aspects of Computer Science (STACS) 2026. To appear.
- [C2] Daniel W. Cranston, Moritz Mühlenthaler, and Benjamin Peyrille*. *A Simple Quadratic Kernel for Token Jumping on Surfaces*. In H. Fernau and P. Kindermann (Eds.), Lecture Notes in Computer Science (Volume 16124) Graph-Theoretic Concepts in Computer Science: 51st International Workshop, WG 2025, Otzenhausen, Germany, June 11–13, 2025, Revised Selected Papers. (pp. 149–161).

- [C1] Anthony Bonato, Daniel W. Cranston, Melissa A. Huggan, Trent Marbach, and Raja Mutharasan. *The Iterated Local Directed Transitivity Model for Social Networks*. International Workshop on Algorithms and Models for the Web-Graph (WAW), pp. 111–123. 2020.

Published Problems

- [P1] Daniel W. Cranston and Douglas B. West. *Bulgarian Solitaire*. Problem 11712, Problems Section of American Mathematical Monthly, June–July 2013.

Doctoral Advising

- May 2024 Reem Mahmoud, PhD. Thesis: Graph Coloring Reconfiguration.
 May 2023 Hudson Lafayette, PhD. Thesis: Selected Problems in Graph Coloring.

Talks

over 185 talks in 9 countries, including 5 Canadian provinces and 31 United States

Conference Plenary Talks

- July 2026 *Reconfiguration of Colorings and List Colorings: Proofs and Conjectures*. AGraCol. Université Paris-Saclay. Orsay, France.
 May 2026 *Reconfiguration of Colorings and List Colorings: Proofs and Conjectures*. 34th Cumberland Conference. U of Auburn.
 Sep. 2019 *Using the Potential Method to Color Near-bipartite Graphs*. WaterColor 2019, Waterloo Coloring Conference. U of Waterloo, Toronto.
 Sep. 2019 *Using the Potential Method to Color Near-bipartite Graphs*. Colourings, Independence and Domination 2019. Piechowice, Poland.
 May 2017 *Edge-coloring Multigraphs*. 29th Cumberland Conference. Vanderbilt.
 Sep. 2015 *Fractional Coloring Planar Graphs and the Plane*. Cycles and Colouring. Slovakia.

Colloquium Presentations

- Mar. 2025 *Reconfiguration of Colorings and List Colorings*. Math Dept. Colloquium. William & Mary.
 Oct. 2023 *An Analogue of Mohar's Conjecture for List-Coloring*. Math Department Colloquium. William & Mary.
 May 2018 *Bootstrap Percolation in Tilings with Regular Polygons*. PME Induction. Randolph-Macon.
 Sep. 2017 *Edge-coloring Graphs and Multigraphs*. Math Department Colloquium. William & Mary.
 Apr. 2015 *Planar graphs are $\frac{9}{2}$ -colorable*. Math Department Colloquium. George Mason.
 Mar. 2015 *Planar graphs are $\frac{9}{2}$ -colorable*. Math Department Colloquium. William & Mary.
 Mar. 2015 *Planar graphs are $\frac{9}{2}$ -colorable*. Applied Math and Stat. Dept. Seminar. Johns Hopkins.
 Feb. 2015 *Planar graphs are $\frac{9}{2}$ -colorable*. Math Department Colloquium. George Washington U.
 Mar. 2013 *A Proof of Bertrand's Postulate*. Undergrad Math Club. Wesleyan U.
 Feb. 2013 *Revolutionaries and Spies*. Math Department Colloquium. Howard U.
 Mar. 2012 *Revolutionaries and Spies*. Math Department Colloquium. William & Mary.
 Oct. 2011 *A Proof of Bertrand's Postulate*. Student Colloquium Series. Louisiana State U.
 Oct. 2011 *Moore Graphs: Beauty is Rare*. Student Colloquium Series. Louisiana State U.
 Mar. 2011 *A Proof of Bertrand's Postulate*. Math Coffee. Davidson College.
 Apr. 2010 *Reducibility and Discharging: An Introduction by Example*. Colloquium. US Naval Academy.
 Apr. 2009 *Coloring and List-coloring of Graphs*. Math Department Colloquium. William & Mary.
 Apr. 2008 *Reducibility and Discharging: Introduction by Example*. CS Colloquium. Rutgers, Camden.

Selected Invited Conference and Seminar Presentations

- Apr. 2026 *Equitably Coloring Planar and Outerplanar Graphs*. Discrete Math Seminar (Virtual). Iowa State U.
- Mar. 2026 *Equitably Coloring Planar and Outerplanar Graphs*. Graph Theory and Combinatorics Seminar (Virtual). U of Illinois.
- Jan. 2026 *Equitably Coloring Planar and Outerplanar Graphs*. Discrete Math Seminar (Virtual). U of Central Florida.
- Oct. 2025 *Rosenfeld Counting: Proper Conflict-free Coloring of Graphs with Large Maximum Degree*. Discrete Applied Math Seminar (Virtual). Illinois Institute of Technology, Chicago.
- May 2025 *Reconfiguration of Colorings and List Colorings: Proofs and Conjectures*. CanaDAM 2025. U of Ottawa.
- May 2025 *Rosenfeld Counting: Proper Conflict-free Coloring of Graphs with Large Maximum Degree*. CanaDAM 2025. U of Ottawa.
- May 2025 *Reconfiguration of Colorings and List Colorings*. Discrete math seminar (Virtual). Iowa State U.
- Mar. 2025 *Reconfiguration of Colorings and List Colorings*. Graph Theory & Combinatorics Seminar (Virtual). U of Illinois.
- Mar. 2025 *Rosenfeld Counting: Proper Conflict-free Coloring of Graphs with Large Maximum Degree*. AMS Special Session. Clemson U.
- Mar. 2025 *Reconfiguration of Colorings and List Colorings*. Discrete Math Seminar. VCU.
- Nov. 2024 *Rosenfeld Counting: Proper Conflict-free Coloring of Graphs with Large Maximum Degree*. Discrete Math Seminar (Virtual). University of Rhode Island.
- Oct. 2024 *A Survey of k -critical Graphs*. Workshop on Albertson's Conjecture and Related Problems. American Institute of Mathematics (Pasadena, CA).
- Nov. 2023 *An Analogue of Mohar's Conjecture for List-Coloring*. Discrete Math Seminar (Virtual). Auburn U.
- Sep. 2023 *Cliques in Squares of Graphs with Maximum Average Degree less than 4*. Discrete Math and CDSOR Seminar (Virtual). Illinois Institute of Technology, Chicago.
- Sep. 2023 *Cliques in Squares of Graphs with Maximum Average Degree less than 4*. AMS Special Session. U of Buffalo.
- Mar. 2023 *An Analogue of Mohar's Conjecture for List-Coloring*. Discrete Math Seminar (Virtual). U of Central Florida.
- Mar. 2023 *Optimally Reconfiguring List Colorings Given Large Lists*. AMS Sectional Mtg. Georgia Tech.
- Mar. 2023 *Optimally Reconfiguring List Colorings Given Large Lists*. Graph Theory and Combinatorics Seminar. Georgia Tech.
- May 2022 *Kempe Equivalent List-Colorings*. Combinatorial Reconfiguration Workshop. BIRS, Banff, Alberta.
- Feb. 2022 *In Most 6-regular Toroidal Graphs All 5-colorings are Kempe Equivalent*. Groups/Graphs, Analysis, and Geometry Seminar (Virtual). William & Mary.
- Feb. 2022 *An Analogue of Mohar's Conjecture for List-Coloring*. UVW Graphs and Combinatorics Seminar (Virtual). U of West Virginia, Morgantown.
- Feb. 2022 *An Analogue of Mohar's Conjecture for List-Coloring*. ISU Discrete Math Seminar (Virtual). Iowa State U, Ames.
- Apr. 2021 *In Most 6-regular Toroidal Graphs All 5-colorings are Kempe Equivalent*. Graphs and Optimisation Seminar (Virtual). LaBRI, France.
- Apr. 2021 *In Most 6-regular Toroidal Graphs All 5-colorings are Kempe Equivalent*. Discrete Math and CDSOR Seminar (Virtual). Illinois Institute of Technology, Chicago.
- Mar. 2021 *In Most 6-regular Toroidal Graphs All 5-colorings are Kempe Equivalent*. Discrete Math Seminar (Virtual). Iowa State U, Ames.

- Nov. 2020 *Vertex Partitions into an Independent Set and a Forest with Each Component Small*. Virtual Discrete Math Colloquium. Center for Mathematical and Computational Sciences in the Institute for Basic Science. Daejeon, South Korea.
- Oct. 2020 *Vertex Partitions into an Independent Set and a Forest with Each Component Small*. SCMS Combinatorics Seminar (Virtual). Shanghai Center for Mathematical Sciences, China.
- Sep. 2020 *Vertex Partitions into an Independent Set and a Forest with Each Component Small*. Graphs and Matroids Seminar (Virtual). U of Waterloo, Toronto.
- Sep. 2020 *Vertex Partitions into an Independent Set and a Forest with Each Component Small*. Discrete Seminar (Virtual). University of Colorado, Denver.
- July 2020 *Vertex Partitions into an Independent Set and a Forest with Each Component Small*. Graphs and Optimisation Seminar (Virtual). LaBRI, France.
- May 2019 *Circular Flows in Planar Graphs*. CanaDAM. Simon Fraser U, Vancouver.
- Mar. 2019 *Circular Flows in Planar Graphs*. (50-minute talk.) Atlanta Lecture Series in Combinatorics & Graph Theory. Georgia Tech, Atlanta.
- Mar. 2019 *Circular Flows in Planar Graphs*. AMS Sectional Meeting. Auburn U, Auburn.
- Oct. 2018 *Coloring Squares of Planar Graphs*. AMS Sectional Meeting. U of Michigan, Ann Arbor.
- Jun. 2018 *Acyclic Edge-coloring of Planar Graphs: Δ Colors Suffice when Δ is Large*. Graph Coloring Minisymposium (organizer). SIAM Disc. Math 2018. Denver, Colorado.
- Apr. 2018 *Bootstrap Percolation Thresholds in Plane Tilings using Regular Polygons*. 59th MIGHTY (Midwest Graph Theory). West Virginia University.
- Mar. 2018 *Coloring Squares of Planar Graphs*. Combinatorics Seminar. George Washington University.
- Jan. 2018 *Acyclic Edge-coloring Planar Graphs*. Int'l Workshop on Graph Theory. Seoul, S. Korea.
- Oct. 2017 *Edge-coloring of Graphs and Multigraphs*. DiscMath Seminar. Illinois State U.
- Sep. 2017 *Acyclic Edge-coloring Planar Graphs*. Graph Theory Seminar. William & Mary.
- Apr. 2017 *List-coloring Claw-free Graphs with $\Delta - 1$ Colors*. AMS Sectional Meeting. IU, Bloomington.
- Oct. 2016 *Edge-coloring Multigraphs*. Graph Coloring Workshop. BIRS, Banff, Alberta.
- Sep. 2016 *Edge-coloring Multigraphs*. Applied Math and Stat. Dept. Seminar. Johns Hopkins.
- Jun. 2016 *Planar Graphs of Girth at least Five are Square $(\Delta + 2)$ -Choosable*. Graph Coloring Minisymposium (organizer). SIAM Disc. Math 2016. Atlanta, Georgia.
- Oct. 2014 *Painting squares with $\Delta^2 - 1$ colors*. Combinatorics Seminar. U of Rhode Island.
- Oct. 2014 *Boundedness of solutions for max-type reciprocal difference equations*. Difference Equations Seminar. U of Rhode Island.
- Jun. 2014 *Graphs with $\chi = \Delta$ have big cliques*. West Fest (honoring 60th birthday of Douglas West). Minneapolis, Minnesota.
- Jun. 2014 *Painting squares with $\Delta^2 - 1$ colors*. Graph Coloring Special Session. SIAM Disc. Math 2014. Minneapolis, Minnesota.
- Apr. 2014 *Graphs with $\chi = \Delta$ have big cliques*. Discrete Math Seminar. Arizona State U.
- Mar. 2014 *Boundedness of Solutions to Reciprocal Max-Type Difference Equations*. AMS Sectional Meeting (Special Session in Difference Equations). U of Maryland, Baltimore.
- Mar. 2014 *Graphs with $\chi = \Delta$ have big cliques*. AMS Sectional Meeting. U of Tennessee, Knoxville.
- Jan. 2014 *Graphs with $\chi = \Delta$ have big cliques*. AMS Special Session in Graph Theory (Co-organizer). Joint Meetings. Baltimore, Maryland.
- Jan. 2014 *Boundedness of Solutions to Reciprocal Max-Type Difference Equations*. AMS Special Session in Difference Equations. Joint Meetings. Baltimore, Maryland.
- Oct. 2013 *Graphs with $\chi = \Delta$ have big cliques*. Discrete Math Days of the Northeast. Wesleyan U.
- Sep. 2013 *Graphs with $\chi = \Delta$ have big cliques*. Discrete Applied Math Seminar. IL Institute of Tech.
- Sep. 2013 *Boundedness of solutions for max-type reciprocal difference equations*. Applied and Computational Math Seminar. George Mason.

- Sep. 2013 *Boundedness of solutions for max-type reciprocal difference equations.* Applied and Computational Math Seminar. VCU.
- Apr. 2013 *Coloring claw-free graphs with $\Delta - 1$ colors.* AMS Sectional Meeting. Iowa St.
- Apr. 2013 *Coloring claw-free graphs with $\Delta - 1$ colors.* East China Normal University. Shanghai, China.
- Apr. 2013 *Star coloring sparse graphs.* Zhejiang Normal University. Jinhua, China.
- Apr. 2013 *Coloring claw-free graphs with $\Delta - 1$ colors.* Zhejiang Normal University. Jinhua, China.
- Mar. 2013 *Revolutionaries and Spies on Graphs.* Applied and Computational Math. Division Seminar. National Inst. of Standards and Tech.
- Feb. 2013 *Coloring claw-free graphs with $\Delta - 1$ colors.* Discrete Math Seminar. U of Delaware.
- Feb. 2013 *Coloring claw-free graphs with $\Delta - 1$ colors.* Combinatorics, Algebra, & Geometry Seminar. George Mason U.
- Jan. 2013 *Coloring claw-free graphs with $\Delta - 1$ colors.* Discrete Math Seminar. West Virginia U.
- Jan. 2013 *Coloring claw-free graphs with $\Delta - 1$ colors.* Graph Theory Seminar. Illinois.
- Jun. 2012 *Conjectures Equivalent to the Borodin-Kostochka Conjecture that seem Weaker.* Graph Coloring Special Session (Organizer). SIAM Disc. Math 2012. Halifax, Nova Scotia.
- Feb. 2012 *Overlap Number of Graphs.* Atlanta Lecture Series in Combinatorics V. Emory.
- Oct. 2011 *Revolutionaries and Spies.* 26th Mini-conference on Discrete Math and Algorithms. Clemson.
- Oct. 2011 *Revolutionaries and Spies.* Combinatorics Seminar. Louisiana State U.
- Sep. 2011 *Linear List-coloring of Sparse Graphs.* AMS Sectional Meeting. Wake Forest.
- Sep. 2011 *List-coloring K_5 -minor-free Graphs.* Combinatorics Seminar. U of South Carolina.
- Aug. 2011 *Crossings, Colorings, and Cliques.* Crossing Numbers Workshop. BIRS, Banff, Alberta.
- June 2011 *Graph Ramsey Theory.* Research Experience for Graduate Students (REGS). Illinois.
- Mar. 2011 *Spies and Revolutionaries.* Special Session on Graph Theory. SIAM Sectional Conference. UNC-Charlotte.
- Nov. 2010 *List Coloring K_5 -minor-free Graphs.* Combinatorics Seminar. George Washington U.
- July 2010 *Maker-Breaker Games: Building a Large Chain in a Poset.* Seminar. West Virginia U.
- July 2010 *Maker-Breaker Games: Building a Large Chain in a Poset.* Research Experience for Graduate Students (REGS). Illinois.
- May 2010 *List Colorings of K_5 -minor-free Graphs with Special List Assignments.* Minisymposium on Graph Theory. AMS Sectional Meeting. New Jersey Institute of Tech.
- Apr. 2010 *Detecting a Machine Failure in a Network: Vertex Identifying Codes.* Discrete Math Seminar. U of Delaware.
- Jan. 2010 *Maker-Breaker Games: Building a Large Chain in a Poset.* SIAM Minisymposium on Graph Theory. Joint Meetings. San Francisco, California.
- Jan. 2010 *Vertex Identifying Codes.* AMS Special Session. Joint Meetings. San Francisco, California.
- Aug. 2009 *Sorting by Cut-and-Paste Moves.* Discrete Math Session. MathFest. Portland, Oregon.
- July 2009 *Injective Colorings of Sparse Graphs.* SIAM Annual Meeting. Denver, Colorado.
- May 2009 *Crossings, Colorings, and Cliques.* Graph Crossing Session. CanaDAM. U of Montreal.
- Apr. 2009 *Crossings, Colorings, and Cliques.* Combinatorics Seminar. Lafayette College.
- Mar. 2009 *Injective Colorings of Sparse Graphs.* Minisymposium on Graph Theory. AMS Sectional Meeting. Illinois.
- Jan. 2009 *Entire- $(\Delta + 4)$ -choosability of Plane Graphs with $\Delta \geq 8$.* SIAM Minisymposium on Graph Theory, I. Joint Meetings, Washington, D.C.
- Oct. 2008 *Colorings, Crossings, and Cliques.* Discrete Math Seminar. Columbia U.
- June 2008 *$(7,2)$ -edge-choosability of 3-regular Graphs.* Minisymposium on Graph Coloring. SIAM Disc. Math. U of Vermont, Burlington.
- May 2008 *How to Compute Jacobians More Efficiently.* 2008 SIAM Optimization: Minisymposium on Graph Coloring for Computation of Sparse Jacobians and Hessians. Boston, Massachusetts.

Mar. 2008	<i>Discharging and Reducibility: An Introduction by Example.</i> Math Seminar. Montclair State.
Mar. 2008	<i>Star Coloring Planar Graphs with High Girth.</i> Discrete Math Seminar. Princeton.
Mar. 2008	<i>Discharging and Reducibility: Introduction by Example.</i> Discrete Math Seminar. Delaware.
Feb. 2008	<i>Regular Bipartite Graphs are Antimagic.</i> Combinatorics Seminar. CUNY.
Nov. 2007	<i>Discharging and Reducibility: An Introduction by Example.</i> Workshop on Algorithms, Combinatorics, and Geometry. U of North Texas.
Oct. 2007	<i>List-coloring the Square of a Subcubic Graph.</i> AMS Regional Meeting. Depaul U.
Mar. 2007	<i>List-coloring the Square of a Subcubic Graph.</i> Seminar. Bell Labs, Murray Hill, N.J.
Mar. 2007	<i>List-coloring the Square of a Subcubic Graph.</i> Seminar. DIMACS Center, Rutgers.
Mar. 2007	<i>Coloring and List-coloring of Graphs.</i> Discrete Math Seminar. Iowa State U.
Oct. 2006	<i>List-coloring the Square of a Subcubic Graph.</i> Graph Theory Seminar. Georgia Tech.
Oct. 2006	<i>List-coloring the Square of a Subcubic Graph.</i> DiscMath Seminar. Illinois State.
Mar. 2006	<i>List Edge-colorability of Planar Graphs with no Kites.</i> DiscMath Seminar. Illinois State.

Teaching Experience (at VCU unless noted)

Spr. 2025	Theory of Computation
Fall 2024	Theory of Computation
Spr. 2024	Theory of Computation
Fall 2023	Theory of Computation
Spr. 2023	Algorithm Analysis
Fall 2022	Theory of Computation
Spr. 2022	Theory of Computation
Fall 2021	Theory of Computation
Spr. 2021	Theory of Computation
Fall 2020	Theory of Computation
Spr. 2020	Linear Algebra (two sections)
Fall 2019	Graph Theory I, Linear Algebra
Spr. 2019	Combinatorics, Linear Algebra
Fall 2018	Graph Coloring, Linear Algebra
Spr. 2018	Linear Algebra, Introduction to Proof
Fall 2017	Graduate Combinatorics, Multivariable Calculus
Spr. 2016	Graph Theory II, Multivariable Calculus
Fall 2015	Linear Algebra (two sections)
Spr. 2015	Combinatorics, Multivariable Calculus, Research Seminar
Fall 2014	Numerical Analysis I, Multivariable Calculus
Spr. 2014	Graph Coloring, Multivariable Calculus, Research Seminar
Fall 2013	Network Models and Graph Theory, Multivariable Calculus
Fall 2012	Linear Algebra, Calculus I (two sections)
Spr. 2012	Linear Algebra, Mathematical Expositions, Graph Theory II (co-taught)
Fall 2011	Linear Algebra, Mathematical Expositions
Spr. 2011	Graph Theory II, Mathematical Expositions
Fall 2010	Linear Algebra, Math Expositions, Graph Theory (co-taught), Problem Seminar (co-taught)
Spr. 2010	Linear Algebra, Discrete Mathematics
Fall 2009	Linear Algebra, Modern Mathematics, Graph Theory (co-taught)
Spr. 2008	Elementary Combinatorics and Probability (Rutgers)
Fall 2000	Calculus I (UIUC)

Fall 1999 Finite Mathematics (UIUC)

Various Mentored 7 VCU students for research seminar and 5 for independent studies

Professional Service

- Refereed over 130 articles for: *American Mathematical Monthly*, *Australasian Journal of Combinatorics*, *Applied Math Letters*, *Ars Combinatoria*, *Ars Mathematica Contemporanea*, *Bulletin of the Iranian Mathematical Society*, *Central European Journal of Mathematics*, *Combinatorica*, *Combinatorics, Probability & Computing*, *Contributions to Discrete Math*, *Discrete Applied Mathematics*, *Discrete Mathematics*, *Discrete Mathematics, Algorithms, and Applications*, *Discussiones Mathematicae Graph Theory*, *Electronic Journal of Combinatorics*, *European Journal of Combinatorics*, *Graph Theory Notes of New York*, *Graphs and Combinatorics*, *Information Processing Letters*, *Information Sciences*, *Innovations in Graph Theory*, *International Mathematics Research Notices*, *Journal of Combinatorial Math and Combinatorial Computing*, *Journal of Combinatorial Optimization*, *Journal of Combinatorial Theory B*, *Journal of Combinatorics*, *Journal of Graph Theory*, *Notices of the AMS*, *SIAM Journal on Discrete Math*, *Theory and Applications of Graphs*, *Utilitas Mathematica*, and a book.
- Reviewed 14 articles (Math Reviews); Reviewed 3 grant proposals (NSA); Panel Reviewer (NSF), 2015.
- Reviewed conference articles for: STOC 2014, WG 2014, EuroComb 2015, SODA 2025, PODC 2026
- Masters students: Bobby Jaeger, May 2015; Coleman Hall, May 2011.
- External Reviewer for Ph.D. Dissertation: Stephen Bradt, Arizona State, February 2026.
- External Reviewer for Ph.D. Dissertation: Hoang La, Université de Montpellier, June 2022.
- External Reviewer for Ph.D. Dissertation: Landon Rabern, Arizona State, April 2013.
- Co-organized VCU Discrete Math Seminar. 2010–2018; Organized 2019–2020.
- Organized Special Sessions on Graph Coloring at Siam Discrete Math: Denver, Colorado, June 2018; Atlanta, Georgia, 2016; Halifax, Nova Scotia, 2012.
- Co-organized Sessions on Graph Theory: Joint Meetings, Baltimore, MD, Jan. 2014; AMS Sectional, Washington, D.C., Mar. 2012; AMS Sectional, Richmond, VA, Nov. 2010.
- Co-organized Special Session on Graph Theory, Joint Math Meetings, Baltimore, MD. Jan 2014.
- Co-organized Graph Theory Special Session at AMS Sectional Meeting in Washington, D.C. Mar. 2012.
- Co-organized Graph Theory Special Session at AMS Sectional Meeting in Richmond, VA. Nov. 2010.
- Co-organized Project NExT panel on “Supervising senior research/capstone projects”. Mathfest 2010.
- Volunteered 14 hours as teaching assistant for *Kids on Campus*. June 2014.
- Volunteered 6 hours consulting for Carle Clinic on scheduling problems. December 2010.
- Mentored or co-mentored 6 undergraduate students at the Rutgers REU. Summer 2008.
- Teaching Assistant for *Math Days* summer math camp for high school students. June 2006.
- Taught mini-course *Mathematical Games* at University High School during Agora Days 2006.
- Coach of math team at University High School. January 2004 to May 2005.
- Teaching Assistant for *SIMUW* math camp for high school students. Summer 2004.